

Unit 4

Functions of single and several variables

① If $f(x)$ is a continuous function $[a, b]$

If $f'(x) > 0 \rightarrow$ Increasing



If $f'(x) < 0 \rightarrow$ decreasing function

If $f'(x) = 0 \rightarrow$ ^{neither} ~~neither~~ ^{increasing} nor decreasing

Function is Stationary

$\rightarrow$ Concave Upward (or) Convex downward

If a portion of a curve on both sides of point lies above the tangent.

It is said to be Concave Upward.



$\rightarrow$ A curve is concave upward in interval

I when $f''(x) > 0$ (Slope is increasing)



$\rightarrow$ Concave downward (or) Convex upward

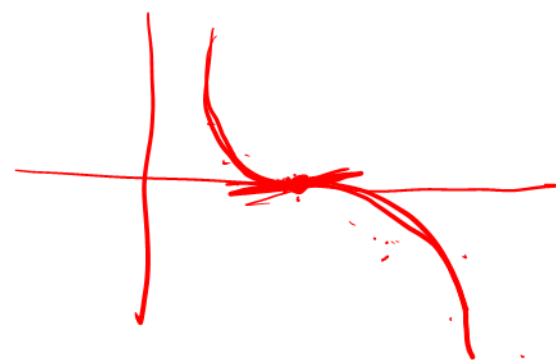
If a portion of a curve on both sides of a point lies below the tangent the curve is said to be concave downward

$\rightarrow$ A curve is concave downward in an interval I

when $f''(x) < 0$ (decreasing slope)

Point of Inflection

If the two portions of curve lie on different sides of a tangent (i.e. curve crosses the tangent)



→ A point on a curve at which curve changes its concavity is called point of inflection.



Test for Point of Inflection

$$f''(x) = 0 \quad \text{but} \quad f'''(x) \neq 0$$

ex:- $y = x^3 + x$ find the points of inflection

$$f'(x) = 3x^2 + 1$$

$$f''(x) = 6x$$

$$f''(x) = 6x = 0$$

$$\boxed{x=0}$$

$$f'''(x) = 6 \neq 0$$

x	< 0	0	> 0
$f(x)$	$-ve$	0	$+ve$
$x^3 + x$			

$x=0$ is point

$(x, f(x))$ - position of inflection
 $(0, 0)$.

ex:- $f(x) = (x-1)^3(x-5)$ find points of Inflection.

$$f'(x) = 3(x-1)^2(x-5) + (x-1)^3(1)$$

$$f'(x) = (x-1)^2 [3(x-5) + x-1] = (x-1)^2 [3x-15+x-1]$$

$$f'(x) = 4(x-1)^2(x-4)$$

$$f''(x) = 4 \cdot 2(x-1)(x-4) + 4(x-1)^2(1)$$

$$= 8(x-1)(x-4) + 4(x-1)^2$$

$$= 4(x-1) [2x-8+x-1]$$

$$= 4(x-1)(3x-9) = 12(x-1)(x-3)$$

$$f''(x) = 12(x-1)(x-3) = 0$$

$$x = 1, 3$$

$$[1, f(1)]$$
$$[3, f(3)]$$

x	$(-\infty, 1)$	$(1, 3)$	$(3, \infty)$
$f''(x)$ $= 12(x-1)(x-3)$	+	-	+

Points of Inflection $(x, f(x))$

$$(1, 0)$$

$$(3, -16)$$

